

ENDERBY ISLAND, New ZealandWMO: **939290**Lat: **50.493S**Lon: **166.296E**Elev: **40**StdP: **100.85**Time Zone: **12.00 (NZL)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.5	1.5	-4.0	2.7	2.0	-3.0	3.0	2.7	19.9	8.7	18.3	8.0	8.0	210	0.983

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.5	13.4	11.9	12.9	11.5	12.4	11.2	12.6	13.0	12.1	12.5	11.7	12.1	9.0	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.5	9.1	12.7	12.0	8.7	12.3	11.5	8.5	11.9	36.0	13.2	34.8	12.5	33.6	12.2	15.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	16.6	15.1	-1.4	14.3	1.0	0.6	-2.2	14.8	-2.8	15.2	-3.4	15.5	-4.1	16.0		
(5)			-2.4	13.4	1.0	0.8	-3.1	13.9	-3.7	14.4	-4.3	14.9	-5.0	15.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	10.1	10.2	9.6	8.6	7.5	6.3	6.1	6.4	6.7	7.1	7.7	9.3
	DBStd	2.21	1.25	1.31	1.42	1.59	2.00	2.04	1.98	1.74	1.77	1.52	1.60	1.45
	HDD10.0	808	14	13	24	46	79	111	121	112	99	91	68	31
	HDD18.3	3789	256	229	271	291	337	361	379	371	349	348	318	280
	CDD10.0	62	17	17	12	4	2	0	0	0	0	0	1	9
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	8.4	8.2	8.2	8.3	8.6	8.5	8.4	8.4	8.3	8.5	8.8	8.4	7.7
(15) Precipitation	PrecAvg	1009	98	80	101	85	94	73	79	69	77	81	75	82
	PrecMax	1203	202	171	162	174	141	181	131	126	153	170	169	154
	PrecMin	807	40	18	53	18	22	1	13	1	12	15	6	26
	PrecStd	122	44	37	27	36	28	38	28	32	36	32	38	28
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	14.1	13.6	12.6	11.7	10.6	10.2	10.2	10.6	11.4	12.1	13.7
		MCWB	12.2	12.8	12.8	11.8	11.0	9.7	9.4	9.2	9.6	9.9	10.3	11.8
	2%	DB	13.3	13.3	12.9	12.0	11.2	10.1	9.6	9.7	10.0	10.6	11.2	12.7
		MCWB	11.5	12.0	12.1	11.3	10.6	9.4	8.9	8.8	9.1	9.4	9.8	11.1
	5%	DB	12.7	12.7	12.3	11.6	10.7	9.6	9.2	9.2	9.6	10.0	10.7	12.1
		MCWB	11.1	11.5	11.5	10.9	10.1	9.0	8.6	8.4	8.8	8.9	9.4	10.7
	10%	DB	12.2	12.3	11.9	11.1	10.3	9.2	8.8	8.8	9.2	9.6	10.2	11.6
		MCWB	10.8	11.2	11.0	10.4	9.7	8.5	8.2	8.1	8.4	8.4	9.1	10.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	13.4	13.2	12.2	11.3	10.2	9.7	9.6	10.1	10.6	11.1	12.5
		MCDB	13.6	13.7	13.3	12.3	11.4	10.4	9.9	9.8	10.2	10.8	11.4	13.1
	2%	WB	12.2	12.7	12.4	11.6	10.8	9.7	9.2	9.1	9.5	9.9	10.4	11.8
		MCDB	12.7	13.0	12.7	11.8	11.0	9.9	9.4	9.4	9.7	10.2	10.8	12.3
	5%	WB	11.7	12.1	11.8	11.1	10.3	9.2	8.8	8.7	9.0	9.3	9.9	11.3
		MCDB	12.2	12.5	12.1	11.4	10.6	9.5	9.1	9.1	9.4	9.7	10.4	11.8
	10%	WB	11.2	11.6	11.3	10.6	9.8	8.6	8.3	8.2	8.6	8.8	9.4	10.7
		MCDB	11.8	12.0	11.7	11.0	10.2	9.1	8.7	8.7	9.0	9.3	10.0	11.3
(35) Mean Daily Temperature Range	5% DB	MDBR	3.6	3.5	3.3	3.2	3.0	3.0	2.9	3.0	3.3	3.8	3.8	3.7
		MCDBR	4.0	3.9	3.4	3.2	3.0	3.0	2.7	2.9	3.2	3.9	3.9	4.0
	5% WB	MCWBR	3.3	3.5	3.4	3.5	3.4	3.4	3.1	3.1	3.1	3.7	3.5	3.3
		MCWBR	3.5	3.4	3.1	3.1	2.9	3.0	2.6	2.8	2.9	3.5	3.4	3.4
(40) Clear-Sky Solar Irradiance	taub		0.346	0.334	0.330	0.319	0.316	0.308	0.311	0.322	0.346	0.336	0.336	0.337
		taud	2.509	2.560	2.571	2.563	2.464	2.386	2.445	2.505	2.448	2.507	2.518	2.537
	Ebn at Noon		963	942	888	802	685	622	678	781	851	929	968	981
		Edh at Noon	106	94	82	67	57	52	56	69	91	98	104	105
(44) All-Sky Solar Radiation	RadAvg		5.37	4.29	2.93	1.74	0.92	0.62	0.79	1.47	2.53	4.06	4.94	5.62
	RadStd		0.32	0.26	0.18	0.10	0.07	0.03	0.06	0.11	0.15	0.16	0.33	0.36

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.36	N/A	+0.66	+0.27	+0.47	+0.57	N/A	-98	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page